Analysis of Clustered Feeding Behavior: From Exponential Decay to Conditional Probabilities

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1 Introduction

In the study of feeding behavior, a recurring challenge is understanding how animals distribute their feeding into *clusters*: do they commonly have small "snacks" (single-pellet events), moderate "meals" (several pellets), or prolonged "feasts" (many pellets in one go)? This document presents a two-stage approach to address these questions:

1. **Exponential Decay Analysis:** We examine the frequency distribution of cluster sizes (number of pellets consumed in one feeding event), checking whether large feeding bouts are exceedingly rare. This stage offers insights into how the *overall* distribution of cluster sizes behaves.

2. **Conditional Probability Analysis:** We then model the likelihood that a cluster of size x pellets grows further to size $x + 1$. This step-by-step transition perspective reveals natural breakpoints in feeding patterns—points where it becomes unlikely for a cluster to continue growing.

Finally, *we use the combined knowledge* from these two stages to propose data-driven boundaries between *Snack*, *Meal*, and *Feast* feeding events.

2 Stage 1: Exponential Decay Analysis

2.1 Model and Fitting Procedure

Let x represent the cluster size (i.e., number of pellets consumed per feeding event), and let y be the observed frequency of events with size x. We posit that

$$y = a\,e^{-bx},$$

where a is a scale parameter (representing the extrapolated frequency at $x = 0$, though $x = 0$ is not observed in practice), and b is the decay constant. A higher value of b indicates a steeper decline in frequency as x increases.

We fit this model by minimizing the sum of squared residuals:

$$\min_{a,b} \sum_{x} \Big[y_{\text{empirical}}(x) \; - \; a\, e^{-bx} \Big]^2,$$

over the range of cluster sizes $1 \le x \le X_{\max}$ that we consider biologically relevant (e.g., up to 10 or 15).

2.2 Results and Interpretation

An example result from our data is:

$$\hat{a} \approx 139.55, \quad \hat{b} \approx 0.28, \quad R^2 \approx 0.9668.$$

This high R^2 indicates that the exponential model explains about 96.68% of the variance in observed cluster frequencies, consistent with the notion that larger clusters drop off rapidly.

2.3 How the Exponential Decay Results Inform the Next Step

Although we do not *numerically* feed $\hat{a}$ or $\hat{b}$ into the conditional probability step, the exponential decay **qualitatively** guides our investigation. Specifically, it confirms that:

- Very large clusters (e.g., above 5 or 6 pellets) are quite rare, suggesting a natural boundary between typical "meals" and unusually large feeding events.
- Smaller clusters (under 5 or 6 pellets) are more common, prompting us to focus on how such clusters evolve incrementally.

Thus, the exponential decay findings *motivate* checking at which cluster size transitions the frequency starts to drop off substantially—leading us to the second step of conditional probabilities.

3 Stage 2: Conditional Probability Analysis

3.1 Definition of $P(x+1 \mid x)$

We define the probability that a cluster of size x will grow by one more pellet, reaching size $x+1$:

$$P(x+1 \mid x) \;=\; \frac{C(x+1)}{C(x)},$$

where $C(x)$ is the total count of clusters of size x. If $C(x) = 0$ for any x, we set $P(x+1 \mid x) = 0$.

3.2 Empirical Estimates

For mice in a specific phase (e.g., non-restricted "NR" phase), meeting certain selection criteria (like "Order 1"), we aggregated cluster sizes and computed:

$$\begin{aligned}
P(2 \mid 1) &= 0.6293,\\
P(3 \mid 2) &= 0.9747,\\
P(4 \mid 3) &= 0.7913,\\
P(5 \mid 4) &= 0.7897,\\
P(6 \mid 5) &= 0.5502,\\
P(7 \mid 6) &= 0.7817,\\
P(8 \mid 7) &= 0.5838,\\
P(9 \mid 8) &= 0.6609,\\
P(10 \mid 9) &= 0.6316.
\end{aligned}$$

This step-wise view shows that some transitions (e.g., $2 \to 3$) are extremely likely (> 0.97), while others (e.g., $5 \to 6$) are barely above 0.55.

3.3 Confidence Intervals for Transition Probabilities

To quantify uncertainty in each $P(x+1 \mid x)$, we use the Wilson score interval at a 95% confidence level. For each x:

$$\hat{p} = \frac{\text{successes}}{\text{trials}},$$

where trials $= C(x)$ and successes $= C(x+1)$. The Wilson method often performs better than the classic Wald interval, especially for proportions near 0 or 1 or with variable sample sizes. An example table of 95% CIs:

Table 1: Observed Conditional Probabilities with 95% Wilson CIs

Size x	$P(x+1 \mid x)$	**CI Lower**	**CI Upper**
1	0.6293	0.6015	0.6562
2	0.9747	0.9609	0.9838
3	0.7913	0.7604	0.8191
4	0.7897	0.7546	0.8209
5	0.5502	0.5044	0.5952
6	0.7817	0.7267	0.8283
7	0.5838	0.5140	0.6503
8	0.6609	0.5704	0.7409
9	0.6316	0.0000	0.0481[1]

[1] Sparse data at size 9 gave an extreme/imprecise interval.

3.4 Relating Conditional Probabilities to the Exponential Decay Findings

Where the exponential decay model revealed a rapid decline for large x, we see in these probabilities that around $x = 5$ or $x = 6$, the chance of adding another pellet drops substantially. Thus, the drop-off from exponential analysis is *reflected* in the conditional probabilities: smaller cluster sizes (1–3 pellets) more reliably grow further, whereas once a cluster is about 5–6 pellets, continuation is less certain. This synergy of results informs our final classification boundaries.

4 Defining Snack, Meal, and Feast

Based on:

1. **Exponential Decay Insight:** Large clusters (above 5 or 6 pellets) are rare, suggesting a boundary point.

2. **Conditional Probability Drop:** Probability from $5 \to 6$ is relatively low (around 0.55), marking a transition zone.

We define:

$$\text{Snack: } x = 1 \quad (1 \text{ pellet}); \quad \text{Meal: } 2 \le x \le 5; \quad \text{Feast: } x > 5.$$

These categories align with observed feeding patterns:

- **Snacks** (single-pellet clusters) rarely evolve further, confirmed by moderate $P(2 \mid 1) \approx 0.63$.

- **Meals** (2–5 pellets) are in a zone where the exponential model predicts moderate frequency, and the conditional probabilities remain high (e.g., > 0.7 for some transitions), suggesting these clusters often continue to some extent.

- **Feasts** (> 5 pellets) are comparatively rare per the decay model, and the conditional probabilities around size 5 or 6 begin to drop, indicating a potential upper boundary for typical meals.

5 Conclusion

Through a sequential analysis that:

1. *Fits an exponential decay* to the frequency of cluster sizes,

2. *Computes conditional probabilities* of cluster growth, and

3. *Examines the synergy* between both results to set classification boundaries,

we have established a data-driven framework to interpret feeding clusters:

- **Snack:** 1 pellet;
- **Meal:** 2–5 pellets;
- **Feast:** more than 5 pellets.

In other words, Stage 1 (Exponential Decay) pointed us to the general *distributional* shape, confirming the rarity of large clusters. Stage 2 (Conditional Probability) then identified specific transitions that are less likely ($5 \to 6$), further supporting a boundary around size 5. Taken together, these analyses yield a biologically motivated partition of feeding events into snack–meal–feast categories.